0.1 Robust Least–Squares Twin SVM for Noisy Binary Classification

In many real-world binary classification problems, datasets are corrupted by outliers and often fail to be linearly separable, which can severely degrade the accuracy of standard margin-based methods. To address both nonseparability and robustness in high-dimensional or overlapping data, we study twin-based SVM models. Next, we investigate how to enhance outlier resilience by replacing the standard squared-slack loss with a rescaled squared-hinge loss, resulting in robust twin-SVM variants. These models preserve a simple linear decision rule while incrementally strengthening their resistance to noise and extreme errors.

Let

$$\mathcal{D}_{+1} = \{(x_i, y_i) \mid x_i \in \mathbb{R}^n,\ y_i = +1\}, \quad \mathcal{D}_{-1} = \{(x_i, y_i) \mid x_i \in \mathbb{R}^n,\ y_i = -1\}.$$

For each class label $k \in \{\pm 1\}$, the Least-Squares Twin SVM fits a hyperplane as follows:

$$\ell_k(x) = b_k^\top x + c_k$$

by solving

$$\min_{b_k \in \mathbb{R}^n, c_k \in \mathbb{R}} \sum_{i \in \mathcal{D}_k} \ell_k(x_i)^2 \ + \ \gamma_k \sum_{i \in \mathcal{D}_{-k}} \big(1 - y_i\, \ell_k(x_i)\big)^2. \tag{LS-TSVM}$$

Solving this problem for each class yields two distinct, non-parallel classifiers, denoted by $\ell_{-1}(x)$ and $\ell_1(x)$, which are then used to assign a label to a new point x according to the rule:

$$\hat{k} \ = \ \arg \min_{k \in \{\pm 1\}} \frac{|\ell_k(x)|}{\|b_k\|_2}.$$

A recent advancement, the Geometric Twin Parametric-Margin SVM (GTPSVM), was proposed in [**rastogi2018robust**], where both hyperplanes are optimized jointly within a unified framework. A least-squares variant of this model, referred to as LS-GTPSVM, is given by:

$$\min_{b_{\pm 1},\, c_{\pm 1}} \tfrac{1}{2} \sum_{k=\pm 1} \left(\|b_k\|_2^2 + c_k^2\right) \ + \ \gamma_1 \sum_{k=\pm 1} \frac{-k}{|\mathcal{D}_{-k}|} \sum_{i \in \mathcal{D}_k} \ell_{-k}(x_i) \ + \ \gamma_2 \sum_{k=\pm 1} \sum_{i \in \mathcal{D}_k} \left(1 - y_i f(x_i)\right)^2, \tag{LS-GTPSVM}$$

where the combined classifier is defined as

$$f(x) := \ell_{-1}(x) + \ell_{+1}(x).$$

At inference time, the label for a new point x is predicted using $\hat{k} = \operatorname{sgn}(f(x))$.

The LS-GTPSVM framework integrates both hyperplanes into a coupled formulation, enhancing the model's robustness and generalization, especially under noisy and high-dimensional settings. The objective includes: (i) a regularization term to control model complexity, (ii) a geometric mean separation term that ensures points from one class lie away from the opposite class's hyperplane, and (iii) a squared margin violation term that symmetrically penalizes classification errors. Unlike LS-TSVM, where each hyperplane is optimized independently, LS-GTPSVM synchronizes the optimization, yielding sharper and more balanced decision boundaries. This integration also allows LS-GTPSVM to better handle outliers and overlapping distributions.

LS-GTPSVM offers several advantages over LS-TSVM. By jointly optimizing both hyperplanes, it enhances robustness and mitigates asymmetry in classification, leading to more balanced separation. The inclusion of regularization terms on both the hyperplane vectors and their biases promotes generalization, particularly in high-dimensional or noisy datasets. Unlike LS-TSVM, which manages margin violations independently for each class, LS-GTPSVM unifies this handling, resulting in a more coherent treatment of outliers and misclassified points. Furthermore, its decision rule, which relies on the average of the two hyperplanes, produces smoother and more stable decision boundaries, ultimately improving performance on unseen data.

The choice of loss function plays a critical role in enhancing model robustness, especially in the presence of outliers. Traditional loss functions, such as the hinge and quadratic loss, are often sensitive to large errors. Specifically, the hinge loss increases linearly with margin violations, resulting in disproportionately high penalties for outliers. Similarly, the quadratic loss grows quadratically with error, further amplifying the influence of outlying observations and increasing the risk of overfitting.

To address this limitation, the rescaled squared hinge loss proposed in [**qi2023novel**] offers a more robust alternative. This loss function grows gradually and saturates for large errors, thereby bounding the influence of extreme margin violations. This bounded nature ensures that even large deviations contribute only modestly to the total loss, thereby improving robustness and reducing overfitting while still preserving sensitivity to moderate errors. The least-squares formulation results in an unconstrained minimization problem, making it well-suited for evaluating the performance of our trust-region algorithm.

Here, we study the least-squares version of this framework as follows:

$$\min_{b_k \in \mathbb{R}^n, c_k \in \mathbb{R}} \sum_{i \in \mathcal{D}_k} \ell_k(x_i)^2 \; + \; \gamma_k \sum_{i \in \mathcal{D}_{-k}} (1 - e^{-\eta(1 - y_i \ell_k(x_i))^2}). \tag{R-LS-TSVM}$$

and analogously, we have:

$$\min_{b_{\pm 1},\, c_{\pm 1}} \frac{1}{2} \sum_{k=\pm 1} \left(\|b_k\|_2^2 + c_k^2 \right) \; + \; \gamma_1 \sum_{k=\pm 1} \frac{-k}{|\mathcal{D}_{-k}|} \sum_{i \in \mathcal{D}_k} \ell_{-k}(x_i) + \gamma_2 \sum_{k=\pm 1} \sum_{i \in \mathcal{D}_k} (1 - e^{-\eta(1 - y_i f(x_i))^2}).,$$
(R-LS-GTPSVM)

where $\eta > 0$ controls the effect of quadratic error.

This sequence of models—(LS-TSVM) through (R-LS-GTPSVM)—moves from *independent* twin hyperplanes (LS-TSVM), to a *coupled* geometric formulation (LS-GTPSVM), and finally to their *outlier-robust* least-squares variants. Each step preserves linear decision functions, adds explicit geometric coupling, and incorporates progressively stronger resistance to noisy or extreme margin violations, yielding a versatile toolbox for challenging binary-classification tasks.

We observe that the rescaled exponential loss rapidly pushes intermediate errors toward 1, making it difficult for the model to distinguish between moderately influential and highly influential points. This limitation motivates us to propose the following rational loss models:

$$\min_{b_k \in \mathbb{R}^n, c_k \in \mathbb{R}} \sum_{i \in \mathcal{D}_k} \ell_k(x_i)^2 \; + \; \gamma_k \sum_{i \in \mathcal{D}_{-k}} \frac{(1 - y_i \ell_k(x_i))^2}{\eta + (1 - y_i \ell_k(x_i))^2}. \tag{R-R-LS-TSVM}$$

And analogously, we have:

$$\min_{b_{\pm 1 \in \mathbb{R}^n},\, c_{\pm 1 \in \mathbb{R}}} \frac{1}{2} \sum_{k=\pm 1} \left(\|b_k\|_2^2 + c_k^2 \right) \; + \; \gamma_1 \sum_{k=\pm 1} \frac{-k}{|\mathcal{D}_{-k}|} \sum_{i \in \mathcal{D}_k} \ell_{-k}(x_i) + \gamma_2 \sum_{k=\pm 1} \sum_{i \in \mathcal{D}_k} \frac{(1 - y_i f(x_i))^2}{\eta + (1 - y_i f(x_i))^2}.$$
(R-R-LS-GTPSVM)